

Electrical wires that can stretch eight times their original length

Researchers at the NC State University have managed to develop electrical wires that can easily stretch eight times their actual capacity. Interestingly, this stretch will not break the wire and is reportedly so smooth that it will not affect the current flow either. Researchers are hopeful that the wire can be used for everything from headphones and phone chargers, and even in electronic textiles.

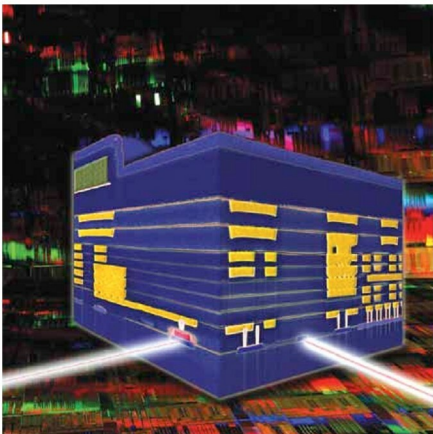
To make the wires, researchers started with a thin tube made of an extremely elastic polymer, then filled the tube with a liquid metal alloy of gallium and indium—an efficient conductor of electricity.

“Previous efforts to create stretchable wires focus on embedding metals or other electrical conductors in elastic polymers, but that creates a trade-off,” said Dr Michael Dickey, an assistant professor of chemical and biomolecular engineering at NC State and co-author of a paper on the research.

“Increasing the amount of metal improves the conductivity of the composite, but diminishes its elasticity. Our approach keeps the materials separate, so you have maximum conductivity without impairing elasticity. In short, our wires are orders of magnitude more stretchable than the most conductive wires, and at least an order of magnitude more conductive than the most stretchable wires currently in the literature,” Dickey added.

Bulb that can turn any surface into touchscreen

What if a table lamp transforms your table into an interactive touch screen? Natan Linder, a PhD student from Fluid Interfaces group at MIT Media Lab, has created an advanced light system named LuminAR that transforms any surface into an interactive touch screen. LuminAR is an augmented reality system that projects images on any surface and detects when anyone touches the projected image.



IBM's silicon nanophotonics technology, where different optical components are placed side-by-side with electrical circuits on a single silicon chip (Courtesy: www.phys.org)

Describing his project at the official site of MIT Media Lab, Linder said, “LuminAR reinvents the traditional incandescent bulb and desk lamp, evolving them into a new category of robotic, digital information devices. The LuminAR bulb combines a pico-projector, camera and wireless computer in a compact form factor. This self-contained system enables users with just-in-time projected information and a gestural user interface. It can be screwed into standard light fixtures everywhere. The LuminAR lamp is an articulated robotic arm, designed to interface with the LuminAR bulb.”

Chip uses light signals to speed up data transfer

In data centres, optical cables are preferred over copper cables as optical cables enable faster data transfer. But the flip side of using these cables is that the data has to be converted into electric signals so that it can be analysed and computed by the processor. To

carry out this entire process, additional equipment are needed.

IBM has made this simpler by developing a low-cost chip that smoothly integrates optical and electrical components side-by-side on the same piece of silicon.

The team working on the project told BBC that using light, rather than the flow of electrons, has two key advantages. One, the data can be sent to longer distances between different parts of a server centre without the risk of losing information. Two, data transfer speeds are faster as light can be used to carry more information at once through cables.

Largest functioning computerised brain

Scientists at the Canada-based University of Waterloo's Centre for Theoretical Neuroscience have created a brain from 2.5 million computer-simulated neurons, which is by far the largest computerised brain. Codenamed as